

Auditory setup

Parts and estimated cost

9 October 2026. Planned recordings in sedated sheep. Prices are in United States dollars; product names link to sellers. “No price” marks a required part whose price needs to be confirmed.

Use the basic setup if tones below 20 kilohertz are enough. For the paper’s 1.4–32 kilohertz range, I suggest pricing the compact setup on page 4. All options assume we can reuse a compatible lab computer and our neural recorder. Labor and recorder connections are extra.

1. Basic setup

Parts: \$346.79, or \$565.79 with a sound-level calibrator.

Computer → sound interface → powered speaker. The measurement microphone connects to the sound interface.

Item	Count	Seller / condition	Cost (\$)
Behringer UMC204HD sound interface	1	eBay / new	97.98
JBL 305P MkII powered speaker	1	Sweetwater / new	99.00
Dayton EMM-6 measurement microphone	1	eBay / new	66.93
Hosa HMIC-010 microphone cable	1	Sweetwater / new	17.99
Hosa HSS-005 audio cable, \$14.95 each	2	Sweetwater / new	29.90
Gator RI-MICTP-FBM microphone stand	1	Sweetwater / new	34.99
Subtotal without calibrator			346.79
Optional: REED R8090 sound-level calibrator	1	DigiKey / new	219.00
Subtotal with calibrator			565.79

- With a borrowed calibrator, allow \$426.79: parts plus an estimated \$80 for delivery and tax.
- Buying the calibrator brings the allowance to \$665.79, including an estimated \$100 reserve. Both exclude the cable or adapter needed to connect timing markers to our neural recorder; the right connection depends on its inputs.

Timing

- Timing error and drift are not known yet. We need to measure when sound reaches the ear and compare that time with the neural recording.
- Repeat timing markers throughout a session to check whether the recordings slowly move out of step. The microphone recording must also be aligned with the neural recording.
- Use the microphone at the ear to check the level of each tone.

For our discussion

- What frequency range do we need?
- What timing inputs does our recorder have?
- Which parts can we borrow, and what budget should we use?

The \$99 speaker price is a limited-time sale. Delivery/tax reserves are estimates. A sound output cannot be assumed compatible with a digital timing input. For scale: a 256-sample buffer at 48,000 samples per second lasts 5.3 milliseconds; this is not measured interface delay. Air adds about three milliseconds per metre, with further speaker/converter delay. References: buffering; sound travel; delay estimates.

2. SONIC setup

Parts with listed prices: \$7,357.58.

This follows the paper’s USB-6361 sound device, SA1 amplifier and MF1 speaker for 1.4–32 kilohertz tones. It includes the paper’s satellite-based time reference, but uses our recorder in place of Connexus.

One sound output channel. Starred items are accessories added to make the chain usable; their exact part numbers are not identified in the paper. National Instruments makes the sound device; Tucker-Davis Technologies makes the speaker and amplifier.

Item	Count	Seller / condition	Cost (\$)
USB-6361 sound generation/recording, 782255-01	1	Newark / new	3,744.42
TimeMachines TM2500C time reference	1	New / backorder	799.99
MF1-M speaker kit	1	Request price	No price
SA1 speaker amplifier	1	Request price	No price
ZB1PS powered enclosure*	1	Request price	No price
781513-01 power supply*	1	Newark / new	123.42
780534-01 locking computer cable*	1	DigiKey / new	69.75
763830-01 United States power cord*	1	Newark / new	51.00
781233-02 cable noise filter*	1	DigiKey / new	38.25
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Sound parts subtotal			4,840.36
Measurement kit (listed on page 4)	1	New amplifier	2,517.22
Subtotal of priced items			7,357.58

Costs not included

- Required parts without confirmed prices: MF1-M speaker, SA1 amplifier and ZB1PS powered enclosure. They are listed above, but their prices must be added to the subtotal.
- Other costs: extra adapters or wiring for the measurement kit and our recorder, network wiring, delivery, tax and labor. The recorder connection depends on its inputs.

Confirm the speaker kit’s included cables and power accessories. The time reference is backordered.

Lower-cost alternative

A seller-new 782255-01 on eBay for \$3,498.95 would save \$245.47. This is an optional replacement for the device above; confirm its accessories and actual condition.

Timing

- SONIC reports 56 milliseconds for recording, filtering and collecting neural data for the decoder. This excludes sensory processing and computer calculation time.
- This is not the error in assigning a time to a sound. The paper does not quantify that error or long-session drift. Our recorder needs its own timing tests.

Buying the sound hardware does not reproduce the paper’s neural-processing delay. The time reference specifies below one microsecond plus network variation, not sound-arrival accuracy. The separate 11-millisecond result came from a one-session analysis. The sound device’s typical free-running clock scale at 25 degrees Celsius is 180 milliseconds per hour versus ideal time; actual drift relative to our recorder is unknown. SONIC recorded awake sheep. The effects of sedation on our responses and timing remain unmeasured.

References: SONIC version 1, pages 5, 7–9 and 11; sound device; time reference; speaker.

3. McGinley lab setup

Parts with listed prices: \$10,068.97.

The supplied slides use a desktop computer and equipment chassis connected to an ED1 driver and ES1 speaker. The list keeps both signal boards; our neural recorder replaces the Neuropixels equipment shown in the slides.

One sound output channel. Most chassis parts are used. The desktop needs a compatible free expansion slot. This route is worth considering if we already have substantial rack hardware.

Item	Count	Seller / condition	Cost (\$)
PXIE-1082 equipment chassis	1	eBay / used	2,650.00
PXIE-6363 signal input/output board	1	eBay / used	2,495.36
PXI-4461 audio module, hybrid 186900-11L	1	eBay / used	699.99
PXIE-8360 chassis connection module	1	eBay / used	493.20
PCIE-8361 computer connection card	1	eBay / listed new	197.00
779500-03 rack cable, 3 metres	1	eBay / listed new	89.85
SHC68-68-EPM cable, 192061C-02	1	eBay / used	89.99
BNC-2090A signal connection panel	1	eBay / used [‡]	613.77
ED1 driver + enclosure/power supply	1	eBay / used	209.06
ES1 speaker + matching cable	1	Request price	No price
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Sound parts subtotal			7,551.75
Measurement kit (listed on page 4)	1	New amplifier	2,517.22
Subtotal of priced items			10,068.97

Costs not included

- Required part without a confirmed price: ES1 speaker with its matching high-voltage cable. It is listed above, but its price must be added to the subtotal.
- Other costs: extra adapters or wiring for the measurement kit and our recorder, any missing chassis power cord, delivery, tax and labor. The recorder connection depends on its inputs.

The ED1 bundle already includes its enclosure and power supply. Check the audio board's actual hybrid connector and calibration before buying.

Timing and frequency range

- The ES1 covers 4–110 kilohertz. It cannot cover the paper's 1.4–4 kilohertz tones.
- Timing error, drift and sound-level accuracy at the ear are unmeasured. The boards' clocks and calibration need to be checked with our recorder.

At 204,800 samples per second, the audio board's flat output extends to 92.1 kilohertz and its output filter adds 0.15625 milliseconds. Speaker and air delay are additional; this is not whole-system timing accuracy. The audio board's internal-clock scale is 72 milliseconds per hour versus ideal time, conditional on calibration and aging. A different reference changes this figure; drift relative to our recorder is unknown. The speaker's raw ± 11 -decibel response variation is not calibrated at-ear accuracy. The connection-panel seller lists no returns. The starred output lead is our addition.

References: supplied diagram; supplied parts slide; audio board, pages 1, 19 and 24; speaker; hybrid variant.

4. Compact setup

Parts with listed prices: \$5,636.47.

This is the option I suggest pricing if we need the full paper frequency range. It uses the computer-connected USB-6361, an SA1 amplifier and an MF2-M speaker, with a microphone near the ear and timing markers recorded with the neural data.

The main list uses a new microphone amplifier. Dagger-marked items make up the \$2,517.22 measurement kit also included on pages 2 and 3. Costs cover the counts listed.

Item	Count	Seller / condition	Cost (\$)
USB-6361 sound generation/recording, 781442-01	1	Newark / new	3,030.42
MF2-M speaker kit	1	Request price	No price
SA1 speaker amplifier	1	Request price	No price
ZB1PS powered enclosure	1	Request price	No price
Earthworks M50 measurement microphone [†]	1	eBay / new	1,299.00
Sound Devices MM-1 microphone amplifier [†]	1	B&H / new	945.00
REED R8090 sound-level calibrator [†]	1	DigiKey / new	219.00
Hosa HMIC-010 microphone cable [†]	1	Sweetwater / new	17.99
Gator RI-MICTP-FBM microphone stand [†]	1	Sweetwater / new	34.99
Panasonic LR6XWA/2SB battery, \$0.62 each [†]	2	DigiKey / new	1.24
Pomona EM4970-36# output lead	1	DigiKey / new	23.68
Hosa PHX-106F-BULK microphone connection lead	1	B&H / new	13.95
Yageo 100,000-ohm resistor, \$0.10 each	2	DigiKey / new	0.20
763830-01 United States power cord	1	Newark / new	51.00
Subtotal of priced items			5,636.47

Costs not included

- MF2-M speaker, SA1 amplifier and ZB1PS powered enclosure: required parts already listed above, without prices. Add their supplier prices to the subtotal.
- Other costs: the recorder connection, delivery, tax and labor.

Device power/cable and speaker stand/cable are included. Microphone wiring and bias returns need review.

Lower-cost alternative

The \$604.50 used MM-1 at B&H replaces the new \$945 amplifier: subtotal \$5,295.97, measurement kit \$2,176.72. The seller lists moderate wear and a 90-day warranty.

Timing and calibration

- Timing error and drift are unknown. One possible test target is to align sound-arrival timestamps with the neural recording within one millisecond after correction. This has not been demonstrated; it is different from SONIC's processing delay.
- The speaker is rated for 1–80 kilohertz and the microphone to 50 kilohertz. The calibrator's 0.5-decibel specification applies at one kilohertz only, not across all our tones.

The setup has not been built or validated. Check repeated markers, sound arrival and converter channel offsets; full-band sound-level accuracy at the ear remains unmeasured. The sound device's clock scale is 180 milliseconds per hour versus ideal time, typical at 25 degrees Celsius; actual recorder-relative drift is unknown. Arduino or a field-programmable gate array would still need the analog sound hardware, acoustic calibration and program development.

References: speaker; microphone; microphone amplifier; calibrator; sound device; detailed comparison.